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Foundation of Machine Learning

Types of Machine Learning -

- (i) Supervised
- (ii) Unsupervised
- (iii) Semi-supervised
- (iv) Reinforcement

Supervised Learning -

- Model is trained on a labelled dataset i.e. each training example has a corresponding output label
- Algorithms: Linear Regression
 - Logistic Regression
 - Decision Tree
 - SVM
 - KNN
 - Random Forest
 - Naive Bayes

Q. How can supervised learning work with * continuous data?

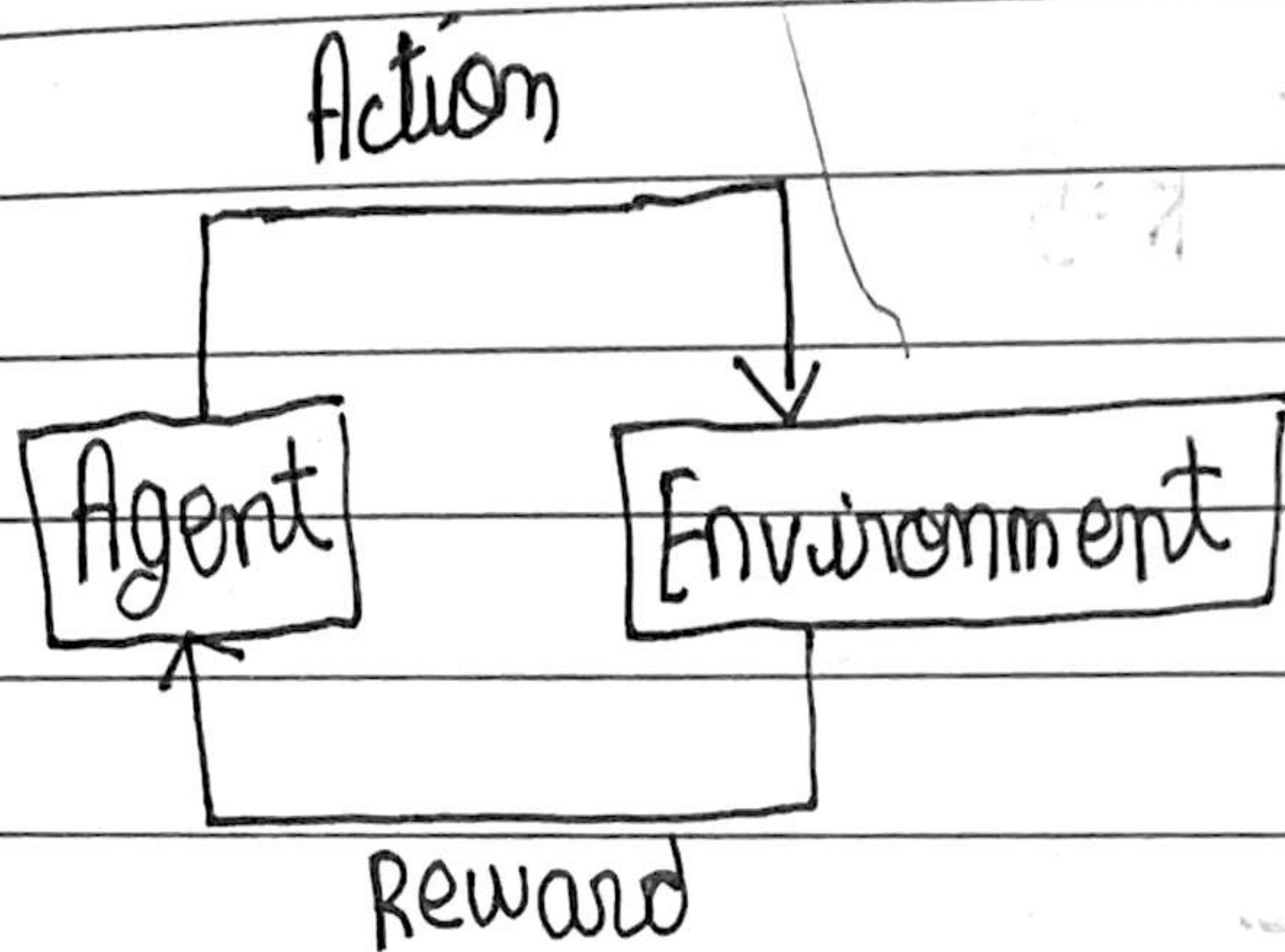
Unsupervised Learning -

- Model is provided with unlabelled data & is tasked with finding hidden patterns & intrinsic structures
- Algorithms: K-means Clustering
 - Hierarchical Clustering
 - DB Scan
 - PCA (Principle Component Analysis)

Reinforcement Learning -

- An agent ~~work~~ learns to make decisions while interacting with the environment receiving rewards or penalties

- Components:
- (i) Agent (Receiver)
 - (ii) Environment (Agent interacts with)
 - (iii) Action (what the Agent can do)
 - (iv) Reward
 - (v) Policy



- Algorithms :
- Q Learning
 - Policy Gradient method

Semi-supervised Learning -

- Used a small amount of labelled data and large amount of unlabelled data.
- Commonly used in situations where labelling is expensive

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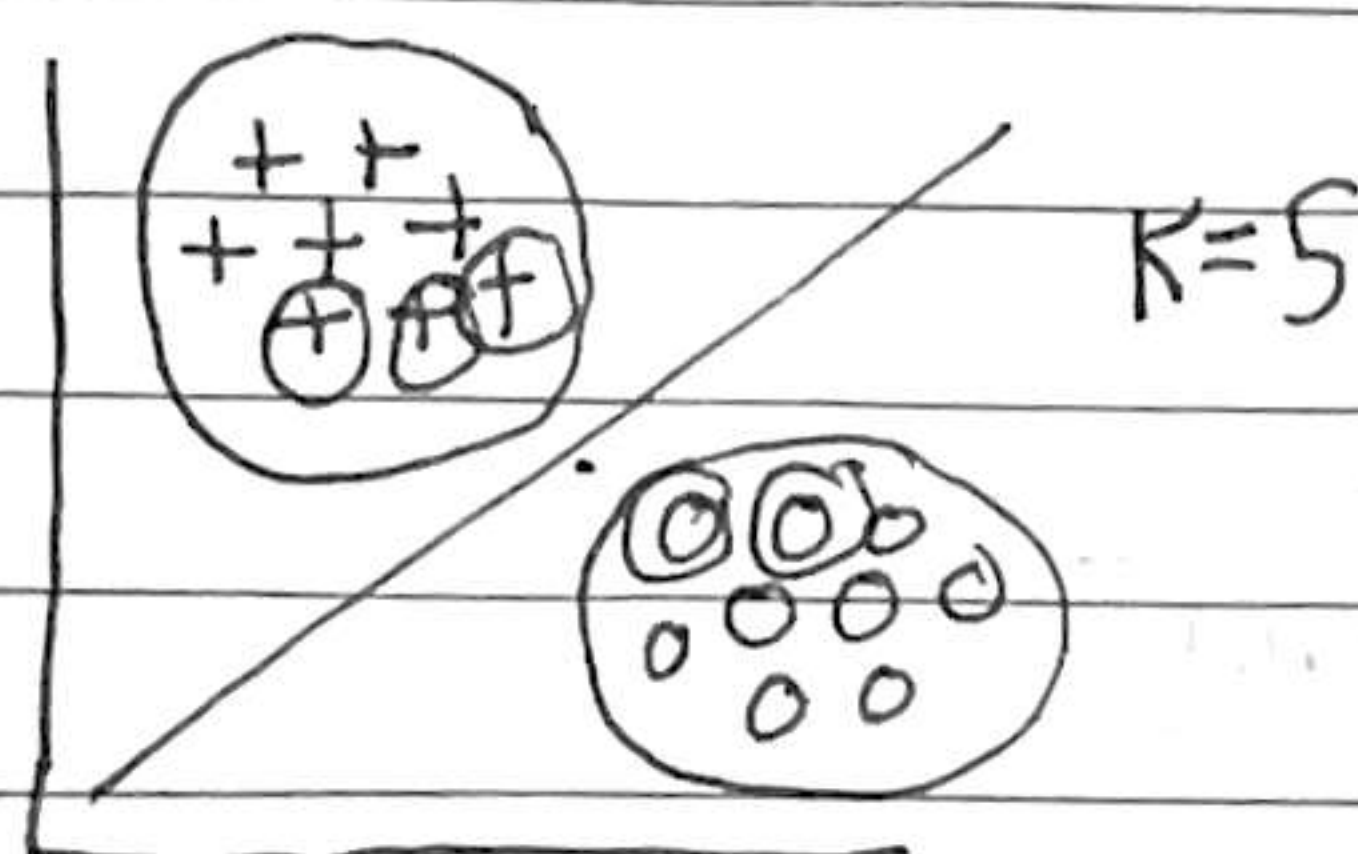
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Handling Missing Data -

1) Imputation

- This involves replacing the missing value into mean, median & mode

2) KNN



Preprocessing



I. Data Cleaning

- (i) Handling missing values
- (ii) Noise/Outlier

II. Data Integration

- Combining of data from multiple sources into a coherent store
- Redundant data occurs when integration of multiple databases is done without taking consideration of object identification & derivable data
- Redundant attributes

*Entity Identification Problem

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Encoding:

Binomial

		Bad	average	Good
1) Label encoding				
2) One hot encoding	0 Bad	0	0	0
3) Ordinal encoding	1 average	0	0	1
	2 Good	0	1	0

→ it is similar to label encoding when we take ~~cat~~ list of categories & convert them into labels

Time Series data -

- A time series is a series of data points indexed in time order)

Most commonly, a time series is a sequence taken at successive equal space points in time.

Thus it is a sequence of discrete time data.

- Data updated in real time required that is up dated in real time requires additional handling and special care for preparing machine learning models

Function for Time-Series data -

- 1) to_date time (—)
- 2) timestamp (—)

⇒ Resampling is done by group by function where we group by based on any column & then apply aggregation functions to resample the data by month, year then there are certain rules for time series

— M for months start
— AS for year start

Supervised Learning -

~~A time~~ Learns from labeled data to predict outcomes.

Goal: Map input features to output labels

Mathematical model: $y = f(x; \theta)$, where θ = learned parameters

Loss: MSE for Regression

Functions: Cross Entropy for classification

Unsupervised Learning -

Learns from unlabeled data to find hidden patterns

Goal: Identify structure in input data without target labels

Mathematical model: optimize structure in data X without Y

Clustering: minimize intra cluster distance objective

Dimensionality: Maximize variance or Reduction objective information retention

Types of Reinforcement Learning -

(a) Model Free vs Model Based -

Model Free: Learns directly from experience

Model Based: Learns a model of the environment

(b) Exploration vs Exploitation -

Exploration: Try new actions to discover rewards

Exploitation: Choose known actions to maximize reward

Key Terms -

(a) Agent: Learner or decision maker

(b) Environment: The external system the agent interacts with

(c) State: A snapshot of the environment at a specific time

(d) Action: Set of possible moves agent can take

- (e) Reward: Feedback from the environment; scalar value
- (f) Policy: Strategy mapping states to actions
- (g) Value Function: Expected cumulative rewards from a state
- (h) Q-Function: Expected cumulative reward from a state action pair

Semi-supervised Learning -

(a) Classification:

- (i) LU Learning: Learning with a small set of labeled examples and a large set of unlabeled examples
- (ii) PU Learning: learning with positive and unlabeled examples (no labeled negative examples)
- (iii) One class learning

(b) Assumption:

- (i) Smoothness assumption: similar inputs have similar outputs
- (ii) Cluster assumption: data points form clusters; same cluster = same label
- (iii) Manifold assumption: data lies on a low dimensional manifold in high dimensional space

=> One way is to use Expectation Maximization (EM)
 The EM algorithm is a popular algorithm for maximum likelihood estimation in problems with missing data
 This ensures that the assumed statistical model, the observed under data is most probable.

It has 2 steps:

Expectation: filling in the missing data

Maximization: calculate a new maximum a posteriori estimate for the parameters

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Feature Selection Techniques -

- 1) Filter methods [Chi Square test
Correlation Analysis] [Pearson
Spearman
- 2) Wrapper methods [Recursive feature elimination
General algorithm
- 3) Embedded method - decision tree

Q. H_0 = Null Hypothesis
 H_1 = Alternate Hypothesis

Watch Action	Y	N	R(Total)
M	140	44	184
F	178	38	216
C(Total)	318	82	400

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

C = Degree of Freedom

$$E = \frac{RT \times CT}{T}$$

Action	Watch	O_i	E_i
M	Y	140	$(184 \times 318) / 400 = 146$
M	N	44	$(184 \times 82) / 400 = 38$
F	Y	178	$(216 \times 318) / 400 = 172$
F	N	38	$(216 \times 82) / 400 = 44$

$(O_i - E_i)$	$(O_i - E_i)^2$	$(O_i - E_i)^2 / E_i$
-6	36	0.246
6	36	0.947
-6	36	0.209
6	36	0.818
		<u>2.220</u>

$[-180, 180]$

$$\alpha = 0.05$$

$$df = (Row - 1) \times (Column - 1)$$

$$= (2 - 1) \times (2 - 1)$$

$$= 1$$

Critical p value for $df = 1$ & $\alpha = 0.05$ is 3.84

Since $3.84 > 2.220$

$\uparrow$ p value

H_0 won't be rejected

1) Pearson coefficient correlation (r)

2) Spearman Correlation

Pearson Correlation

r

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

H_0 = that r does not differ

Pearson correlation

- It measures the relationships between 2 continuous variables & assumptions that we consider both variables as normally distributed.
- It can be applied on Numeric data.
- Highly sensitive to outliers
- Range = $[-1, 1]$

-1 =

0 = No linear correlation

1 = Perfect Positive linear correlation

Spearman Rank correlation -

- Measures monotonic relationship using ranked values.
- Assumptions that there is no strict distribution
- Works on both nominal & ordinal data
- Not as sensitive to outliers
- Range = $[-1, 1]$

Q.	X	Y	$(x_i - \bar{x})$	$(y_i - \bar{y})$	$(y_i - \bar{y})$
	12	6	-12		-9
	16	9	-8		-6
	20	12	-4		-3
	24	15	0		0
	28	18	4		3
	32	21	8		6
	36	24	12		9
	168	105			

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

$$\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}$$

$$\sqrt{448 \cdot 288}$$

$$\sqrt{12896}$$

$$336$$

$$336$$

$\bar{x} = 24$	$\bar{y} = 15$	$(x_i - \bar{x})$	$(y_i - \bar{y})$	$(x_i - \bar{x})$	$(y_i - \bar{y})$
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108

144

81

48

64

36

12

16

9

0

0

0

12

16

9

48

64

36

108

144

81

$$= \frac{336}{336} = 1$$

$$336$$

Q In an art comp. 2 judges awarded following ranks:

X Y $D = (R_1 - R_2)$ D^2

1 6 -5 25

2 2 0 0

3 9 -6 36

4 7 -3 9

5 1 4 16

6 4 2 4

7 8 -1 1

8 3 5 25

9 10 -1 1

10 5 5 25

$$k_s = 1 - \frac{6 \times 142}{10(99)}$$

$$= 1 - 852$$

$$990$$

$$= 1 - 80.860$$

$$= 0.14$$

$$142$$

M A R₁ R₂ $(R_1 - R_2)$ $(R_1 - R_2)^2$

14 4 -5 7 7 -2 4

15 12 4 1 1 3 9

17 8 2 4 4 -2 4

12 10 6 2 2 4 16

16 2 3 9 9 -6 36

11 5 7 6 6 1 1

18 9 1 3 3 -2 4

9 3 9 8 8 1 1

10 7 8 5 5 3 9

$$84$$

$$k_s = 1 - \frac{6 \times 84}{9(83)}$$

$$= 1 - \frac{504}{567}$$

$$= 1 - 0.88$$

$$= 0.12$$

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Wrapper methods -

- techniques in ML, used to select important features (variables) from a dataset by actually training and evaluating models to see which feature combinations work best.

1) Feature Selection:

- Starts with no features in your model
- Add features one by one
- Each time, choose the feature that boosts model performance the most.
- Stop when adding more features doesn't improve things further
- Building a sandwich. We start with an empty plate & at each step, you add one ingredient.

2) Backward elimination:

- Start with all features included
- Remove least useful feature (one whose removal improves performance)
- Keep removing features one at a time, testing each time
- Stop when removing any more feature makes the model perform worse
- Fully loaded pizza, with every possible topping.

Then one by one, you remove the topping that affects the taste the least or isn't needed. After each removal, you taste again & repeat

3) Recursive Feature elimination (RFE)

- Start with all features
- Repeatedly build a model & remove the least important feature (as ranked by the model)
- Continue the process until a set number of features is left or until removing more features hurt performance

Embedded Methods:

- Lasso Regularization
- Decision Tree
- Random Forest

A supervised learning model that splits data into branches based on feature values; easy to interpret

→ An ensemble / collection of multiple decision trees

1) Lasso Regularization

Reducing Reduces overfitting, ↑ accuracy

Regularization is a technique used in ML to prevent overfitting by adding penalty to the model loss function.

This discourages the model fitting too closely to the training data and encourages it to generalize better on unseen data.

Imputation technique

- Backward and forward fill

- Encoding numerical

- Feature scaling

min-max
Z score

- Time Series

Q. Diff between Normalization & standardization & its application

Binning Method

- Handling Outliers

- Reducing noise

- Improving interpretability

Equal width Binning

The range of data is divided into same size of interval or width

$$\text{Bin width} = \frac{\text{max value} - \text{min value}}{\text{no. of bins}}$$

Q. 2, 4, 7, 10, 12, 20, 15, 18

$$\frac{20-2}{4} = \frac{18-18}{4} = 4.5$$

(i) [2-6.5] 2, 4

(ii) [6.5-11] 7, 10

(iii) [11-15.5] 12, 15

(iv) [15.5-20] 18, 20

* Equal frequency Binning

Known as quantile. Each bin contains same no. of data points. Values are sorted & divided so that each bin has equal frequency.

Q. 5, 8, 12, 15, 18, 22, 25, 28, 30, 35, 40, 50

Divide data into 3 bins

$$\text{Total} = 12$$

[5, 8, 12, 15]

[18, 22, 25, 28]

[30, 35, 40, 50]

* But if there are odd no. of data points & diff between data point is large then place it into next bin

→ Transforms correlated variables into a smaller set of uncorrelated variables called Principle components

Principle component Analysis

1) Calculate mean of every feature

2) Calculate covariance matrix

$$COV = \begin{bmatrix} \text{var}(x_1) & \text{Cor}(x_1, x_2) \\ \text{Cor}(x_2, x_1) & \text{var}(x_2) \end{bmatrix}$$

3) Calculate eigen values & eigen vectors

4) Choose PCA

5) Project data into Principle component

X_1 X_2

5 10

3 7

10 15

2 4

5 9

$$COV = \begin{bmatrix} 12.67 & 16.33 \\ 16.33 & 22 \end{bmatrix}$$

$$COV = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

$$\begin{bmatrix} 12.67 & 16.33 \\ 16.33 & 22 \end{bmatrix}$$

~~Graphs~~~~Category~~# ~~Min-max Normal~~

Power Transformation -

- Adjusts data distribution using mathematical function like sqrt, log, or Box Cox.

(i) Square root transformation -

$$y = \sqrt{x}$$

Used for reducing right skewness, especially with count data

(ii) Log transformation -

$$y = \log(x)$$

Helps stabilize variance & normalize data, but only for positive values

(iii) Reciprocal transformation -

$$y = \log \frac{1}{x}$$

Strongly reduces the impact of large values.

(iv) Box Cox transformation -

for $(x > 0)$:

$$y = \frac{x^\lambda - 1}{\lambda} \text{ if } \lambda \neq 0 \text{ (Usually 1.5)}$$

$$y = \log(x) \text{ if } \lambda = 0$$

Box Cox finds the best power (λ) to make data more normal (less skewed)

* We use power transformations to help make data more symmetric (less skewed), stabilize variance and make relationships between variables more linear

Normalization vs Standardization

Aspect	Normalization	Standardization
1) Output Range	Fixed	No fixed range
2) Based on	Min & max values	Mean & SD
3) Goal	Rescale data to a specific range	Center data around the mean (0) & scale to unit variance (SD=1)
4) Formula	$x' = \frac{x - \min}{\max - \min}$	$Z = \frac{x - \mu}{\sigma}$
5) Sensitive to outliers	Yes	Less sensitive
6) Handles outliers	Poorly	Better
7) Preserves distribution	No	Yes
8) Common Algo	KNN, neural networks	PCA, regression, clustering
9) Effect on shape	Can distort original distribution	Retains original distribution shape
10) When to use	When scale matters	When distribution matters

Data transformation -

- Process of converting data into a suitable format for analysis or modeling

- Types:

(a) Scaling transformations:

└ min max scaling

└ standardization (z score)

(b) Power transformations:

└ sqrt, log, reciprocal

└ box cox

(c) Normalization:

└ L1 Normalization = scales so sum of absolute values is 1

└ L2 Normalization = scales so sum of squared values is 1

(d) Categorical Transformations / encoding:

└ One Hot encoding = converts categories to binary vector

└ Label encoding = assigns integers to categories

└ Ordinal encoding = like label encoding, but sequence matters

(e) Discretization / binning:

└ Binning = converts continuous data into discrete intervals

Dimensionality Reduction:

- Process of reducing the no. of features (variables) in a dataset while retaining imp information,

- Helps avoid redundancy & simplifies visualization

- Makes classification problems easier, helps manage datasets with many features & converts high dimensional data into simpler forms

Components of Dimensionality Reduction:

- (i) Feature selection = choosing a subset of imp features (via Filter, wrapper, or embedded methods)
- (ii) Feature extraction = Transforming data into a lower dimensional space

Methods:

- (i) Principle Component Analysis (PCA)
- (ii) Linear Discriminant Analysis (LDA)
- (iii) Generalized Discriminant Analysis (GDA)

Advantages vs Disadvantages:

Advantages

- 1) Reduces storage space
- 2) Lowers computation time
- 3) Removes redundant features

Disadvantages

- 1) May cause some data loss
- 2) PCA only captures linear correlations
- 3) Fails if mean & covariance don't define data well

Applications:

- 1) Face Recognition
- 2) Gene expression Analysis

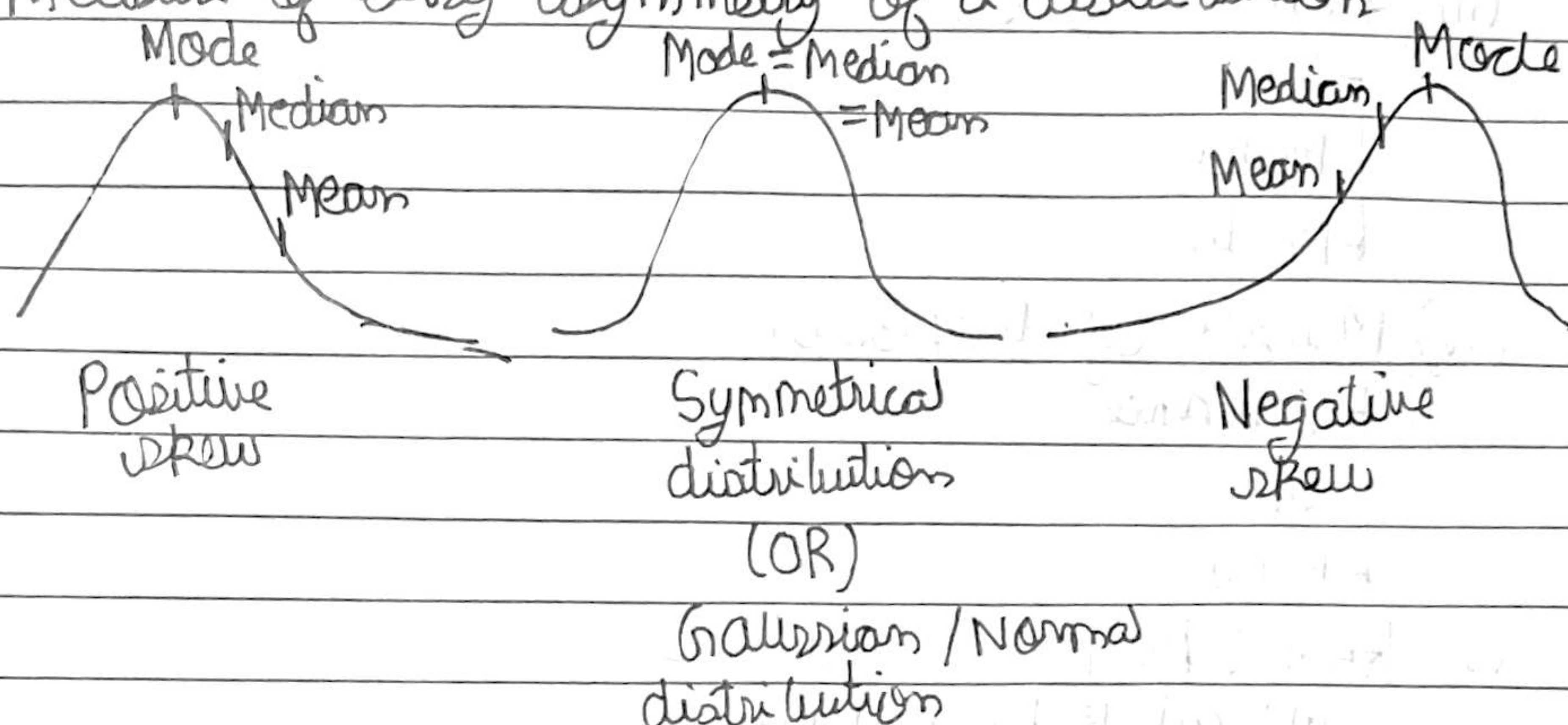
Filter vs Wrapper vs Embedded Methods:

Method	Description	Advantages	Disadvantages
1) Filter	Selects features using statistical measures. Does not depend on learning algo.	Fast, simple, low cost, avoids overfitting	Ignores feature interaction; may miss best subset
2) Wrapper	Uses a predictive model to evaluate subset of features by training repeatedly	Considers feature interactions; usually more accurate	Computationally expensive; risk of overfitting

3) Embedded	Combines filter + wrapper; selects features during model training	Balance of accuracy & efficiency; iterative & robust	Still also dependent more complex to implement
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Skewness

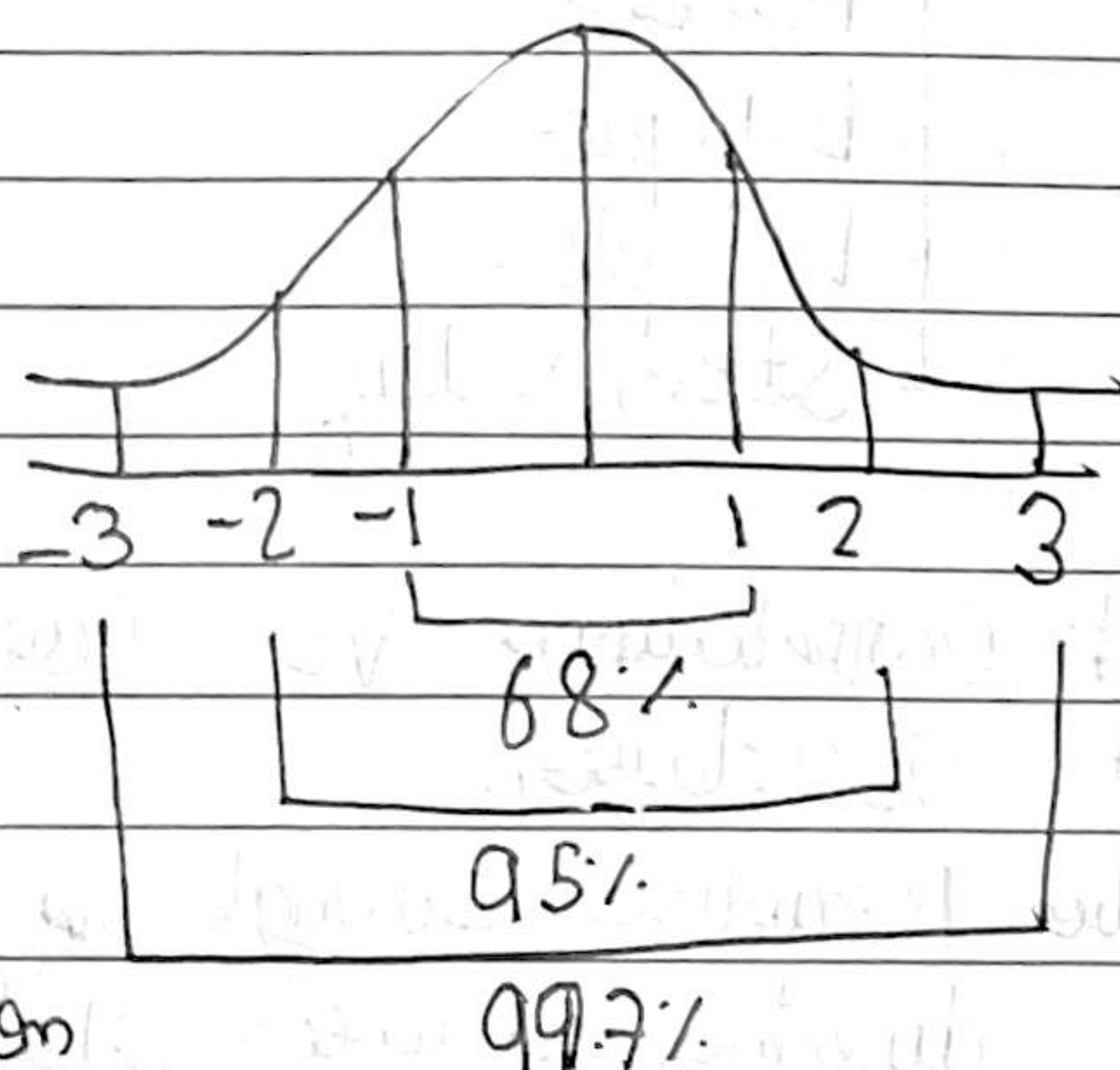
- Measure of ~~only~~ asymmetry of a distribution



Empirical Rule -

(68-95-99.7% Rule)

- 68% data within $\pm 1\sigma$
- 95% data within $\pm 2\sigma$
- 99.7% data within $\pm 3\sigma$
- Q-Q plot checks normality



Standard Normal Distribution (SNR) -

- Special normal distribution where mean = 0 & SD = 1
- Any normal distribution can be standardized by converting values into Z scores

Exploratory Data Analysis (EDA) -

Process of analyzing & summarizing datasets using statistical methods & visualizations to discover patterns, detect anomalies, test hypothesis, etc

(i) Collect data & gain domain knowledge

(ii) Confirm data types & probabilities

(iii) Measure of Central tendency:

- Mean

- Median

- Mode

(iv) Measure of Dispersion:

- Variance

- SD

- Range

(v) ~~Kurt~~ Skewness & Kurtosis

(vi) Graphical Representation:

- Histogram

- Boxplot

- Bar plot

- Dot plot

- Stem & leaf

Correlation vs Covariance -

Aspect

Correlation

Covariance

1) Definition

Measures strength & direction of linear relation

Measures degrees of joint variability

2) Range

-1 to +1

(- ∞ , + ∞)

3) Scale

No

Yes

dependent

4) Interpretation

+1 = perfect positive

-1 = perfect negative

0 = none

Positive = move together

Negative = move opposite

5) Use case

For comparing relationships across datasets

For magnitude of relationship between in original units

Data Structures in Pandas -

- (i) Series = 1D labeled array (like a column)
- (ii) DataFrame = 2D labeled, tabular data (rows + columns)

Overfitting VS Underfitting -

Aspect	Overfitting	Underfitting
1) Definition	Model learns noise + details, performs well on train but poorly on test	Model is too simple, fails to capture data patterns
2) Cause	Too complex model, too many features	Too simple model, too few features, high bias
3) Training error	Very low	High
4) Test Error	High (poor generalization)	High (poor accuracy)
5) Solution	Simplify model, use regularization, more data	Use complex model, add features, reduce bias

Matplotlib -

- 1) Low level library for 2D plotting in Python
- 2) Provides full control over plots (axes, labels, styles)

Pyplot -

- 1) Module in Matplotlib (imported as plt)
- 2) Simplifies plot creation with easy functions like plot(), show()

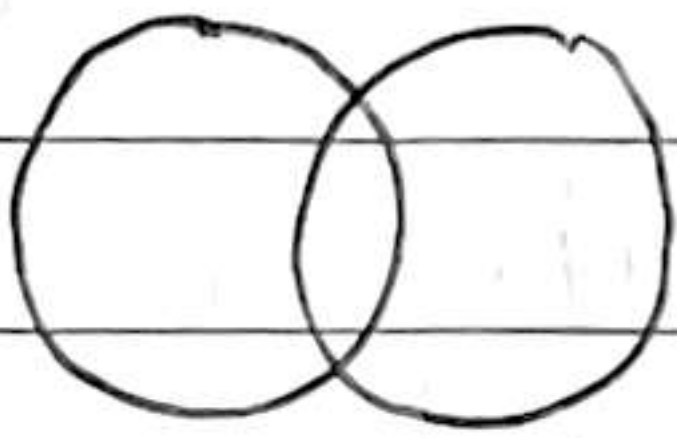


Seaborn -

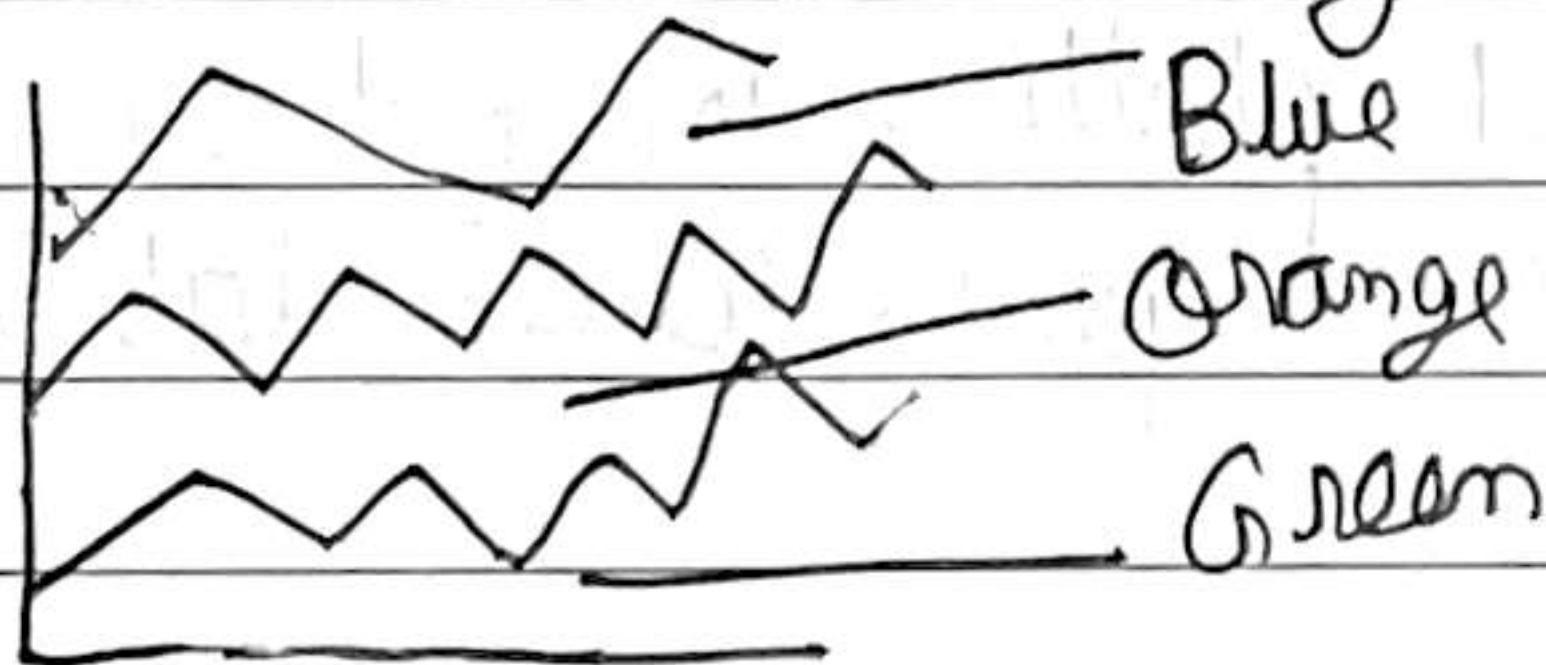
- 1) Built on top of matplotlib
- 2) High level interface for statistical graphics with better aesthetics

Graphs & Charts →

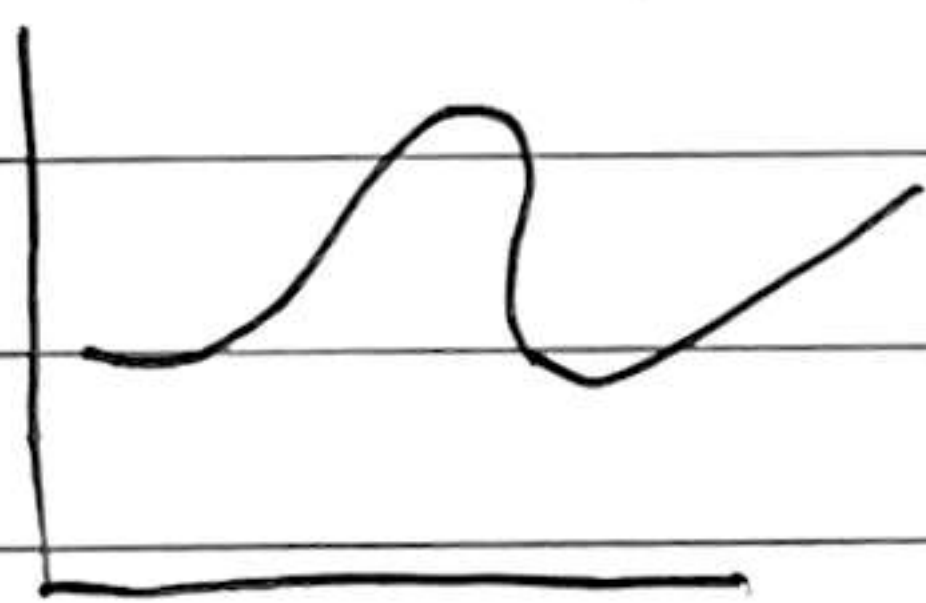
- 1) Line Graph:
Shows trends or changes over time
- 2) Bar chart:
Compares categories (nominal/ordinal data)
- 3) Pie chart:
Shows proportions/percentages of a whole
- 4) Histogram:
Displays frequency distribution of continuous data
- 5) Scatter plot:
Shows relationships/correlation between 2 variables
- 6) Venn diagram:
Visualizes logical relationships/overlaps between groups



- 7) Area chart:
Shows trends with magnitude / filled under line



- 8) Spline chart:
Like line chart but with smooth curves



- 9) Box & Whisker:
Shows data spread, quartiles, median & outliers

Time Series Data -

- A sequence of data points collected at successive, equally spaced time intervals (eg- stock prices, monthly ~~prices~~ sales, weather records)
- Used to study trends, seasonality, & patterns over time

Time Series Analysis -

- Study of data points collected over time at equal intervals to identify trends
- Components:
 - (i) Level: Baseline or average value around which fluctuation occur
 - (ii) Trend: Long term upward or downward movement
 - (iii) Seasonality: Repeating short term patterns
 - (iv) Cyclic: Long term oscillations
 - (v) Irregular (Noise): Random, unpredictable variations
- A stationary series has constant mean, variance & autocovariance over time
- Characteristics:
 - (i) Data is time ordered
 - (ii) Observations are dependent
 - (iii) Collected at regular intervals
- Application:
 - Stock market prediction
 - Weather prediction
 - Sales & demand forecasting
 - Economics

Multi collinearity -

- Occurs when 2 or more independent variables in a dataset are highly correlated, making it hard to determine their individual effect on dependent variable
- Methods to detect:

(i) Correlation matrix = checks correlation coefficients between predictors (high > 0.8 indicates multi collinearity)

(ii) Variance Inflation Factor (VIF):

$VIF > 10$ usually signal multi collinearity